PSYLIQ Investigative Data Analysis Internship
Project 1: Scam Execution & Victim Profiling
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Executive Briefing

Cyber-enabled fraud in the United States has evolved into a structured and adaptive ecosystem that exploits human behavior, digital platforms, and financial anonymity. This report investigates who is targeted, how scams are executed, and how effectively institutions respond, using non-financial open-source intelligence.

The objective of this analysis is to transform raw scam data into actionable defensive intelligence that can support Psyliq's real-time intervention strategies and long-term security planning.

Tools & Data Sources

Analytical Tools

- **Python (Pandas, Matplotlib)**
- **Microsoft Excel**

Data Sources

- **FBI IC3 Annual Reports (2020–2024)**
- **FTC Consumer Sentinel Network**
- **U.S. Department of Justice Press Releases**
- **CISA and FBI Public Advisories**
- **Verified victim narratives and investigative reports**

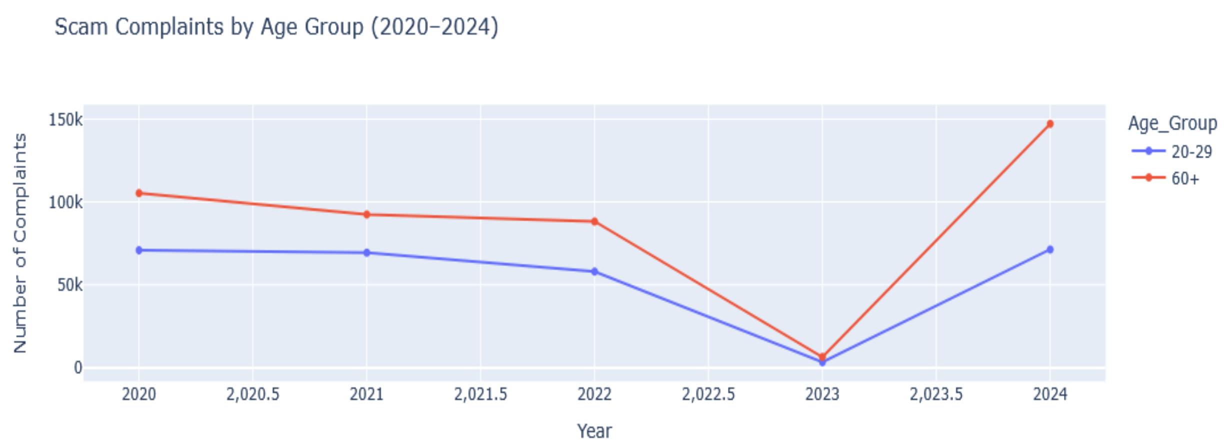
1.1 Victim Demographics & Geographical Hotspots

Methodology

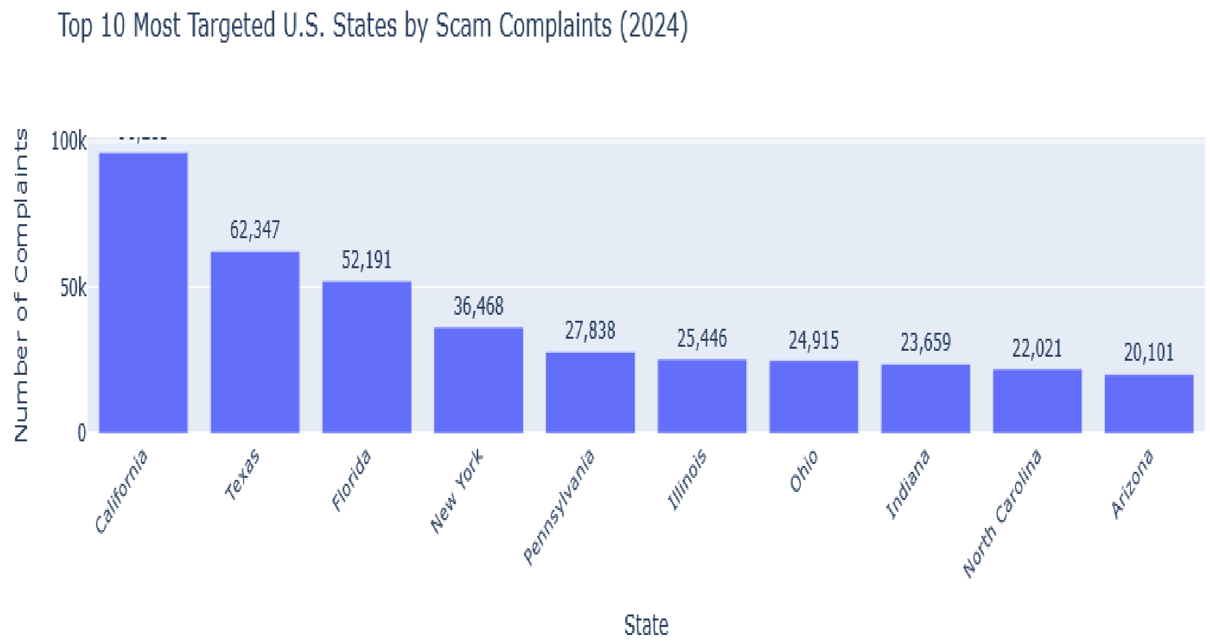
Victim complaint data was aggregated by age group and U.S. state across five years. Data cleaning and standardization were

performed before visualizing trends using time-series analysis and geographic mapping.

📌 Figure 1: Time-Series Chart – Complaint Volume (Age 20–29 vs 60+, 2020–2024)



📌 Figure 2: Bar Chart – Top 10 Targeted States by Complaint Volume (2024)



Key Findings

Victims aged 60+ show the strongest growth in complaint volume and remain highly targeted. Adults aged 20–29 also report consistently high exposure, particularly through online platforms. (The dataset used here does not include loss-per-incident fields, so financial-loss comparisons by age are not inferred.)

Observation

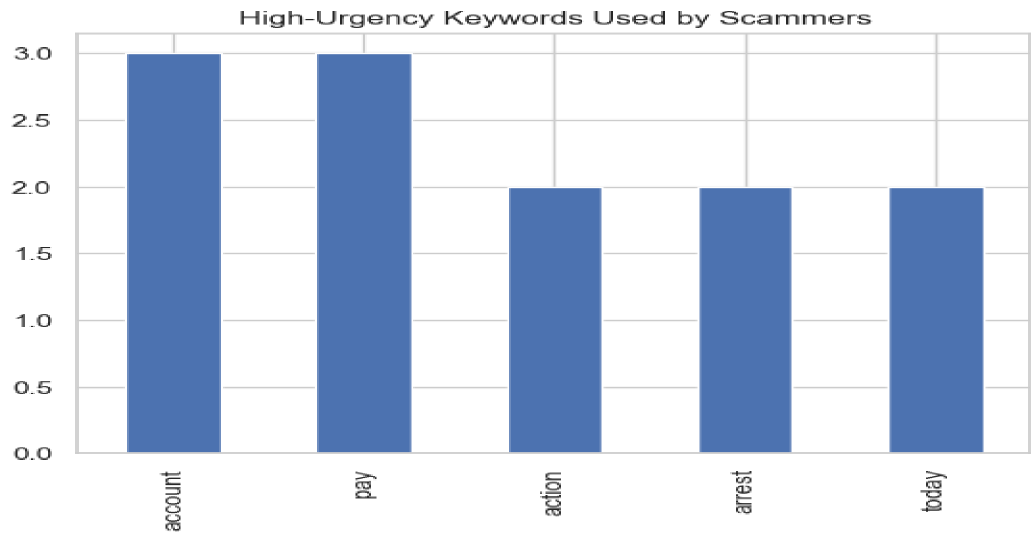
One unexpected trend is the rapid rise in scam victimization among younger adults (20–29). This challenges the common assumption that digital familiarity equates to digital safety. Despite being more comfortable with online platforms, younger users are increasingly exposed to platform-based impersonation and marketplace scams, suggesting that scammers are now optimizing their tactics for speed and trust rather than technical complexity.

1.2 Scammer Tactics & Intervention Analysis

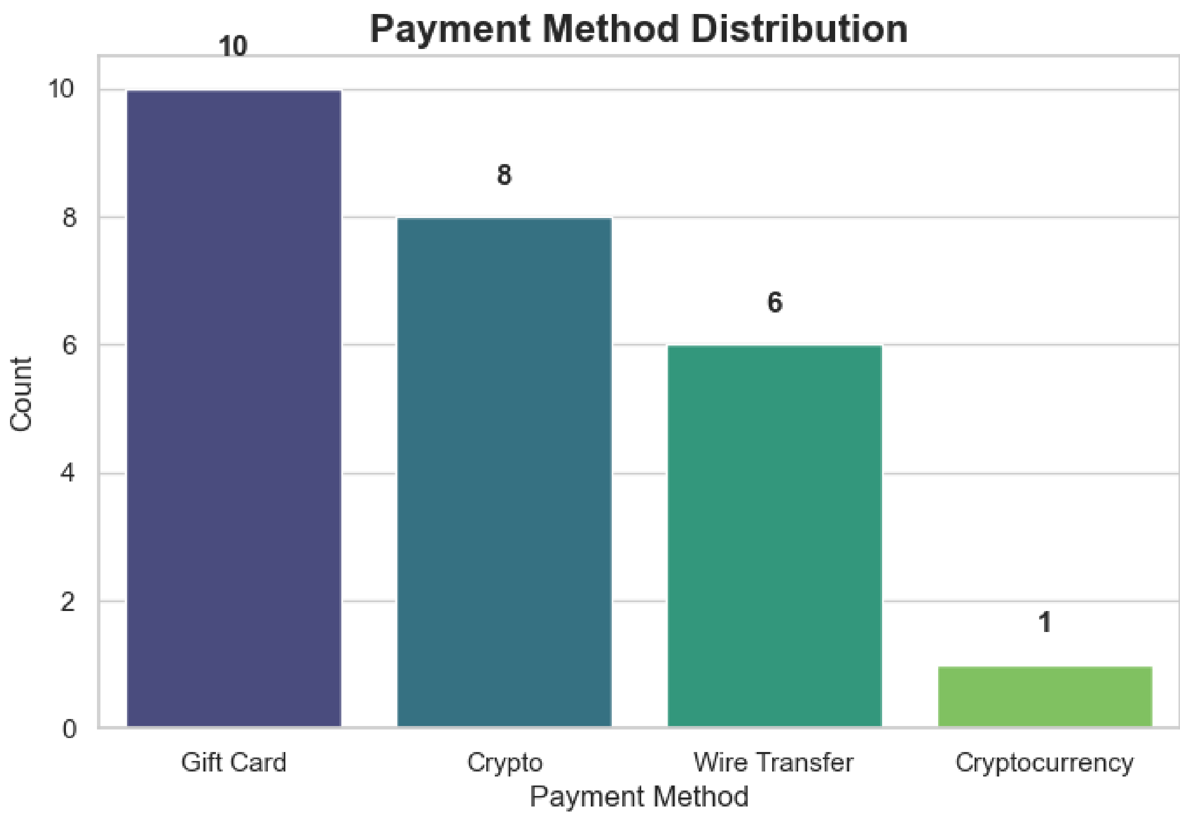
Methodology

Scammer narratives were extracted from victim stories and government advisories. Text analysis techniques were applied to identify urgency-driven language, commonly used platforms, and demanded payment methods.

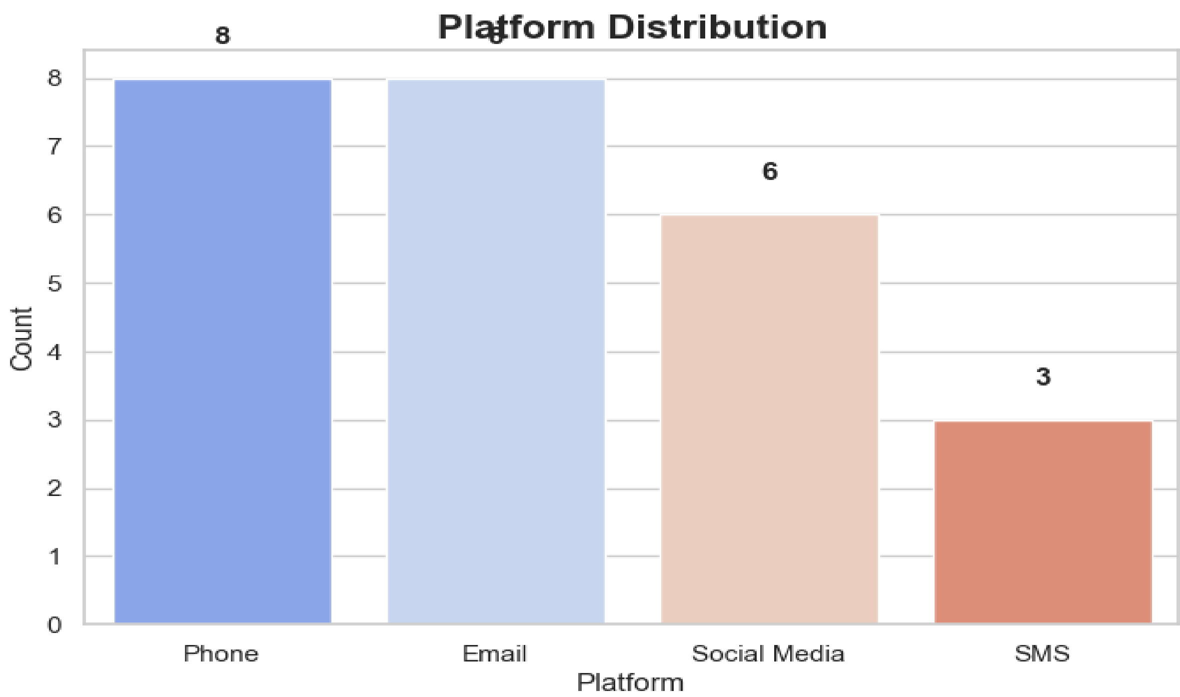
📌 Figure 3: High-Urgency Keyword Frequency



📌 Figure 4: Payment Methods Demanded by Scammers



📌 **Figure 5: Scam Platform Distribution**



Key Findings

The most common high-urgency keywords include:

- **Arrest**
- **Account**
- **Pay**
- **Today**
- **Action**

Scams consistently rely on authority impersonation and fear-based escalation to pressure victims into rapid decision-making.

Typical Imposter Scam Script

1. **Authority impersonation (bank, police, tech support)**

2. **Fear creation (account suspension or arrest threat)**
3. **Urgency escalation**
4. **Victim isolation**
5. **Payment extraction**
6. **Disappearance**

Observation

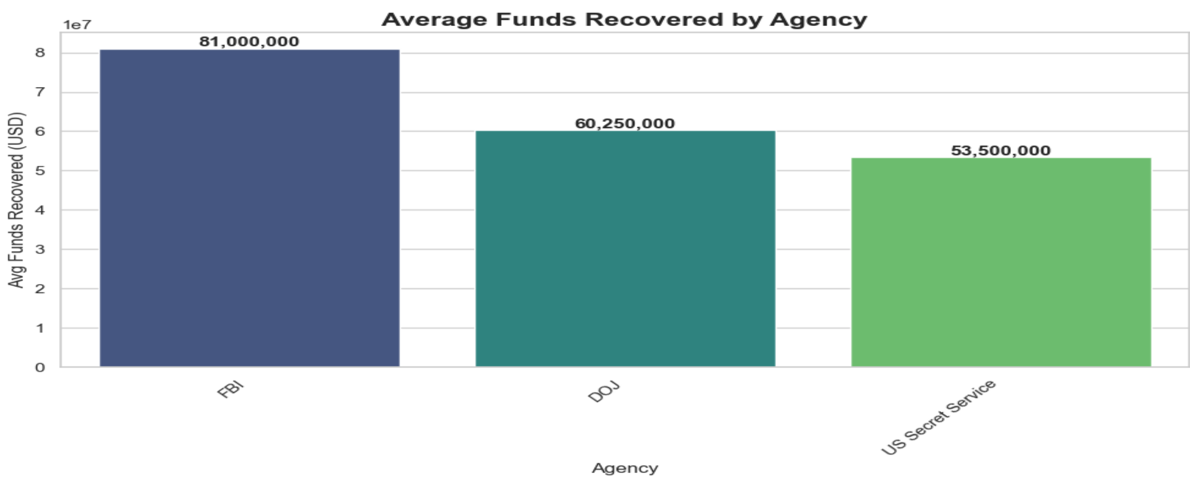
Across the sampled victim narratives, gift cards and cryptocurrency are among the most frequently demanded payment methods, with wire transfers also common—highlighting scammers' preference for fast, hard-to-reverse transactions.

1.3 Government Response & Future Defense

Methodology

Major DOJ enforcement actions were structured into a relational dataset. Python-based queries were used to calculate average estimated funds recovered per agency.

📌 **Figure 6: Average Estimated Funds Recovered per Agency**



Key Findings

Agencies with specialized cybercrime units recover higher average funds per action. However, most enforcement efforts occur after significant financial damage has already been done.

Observation

While government agencies have successfully disrupted several large-scale scam operations, most actions occur after significant financial damage has already been done. Current enforcement strategies focus on arrests and fund recovery rather than early detection. This creates a critical gap where victims remain unprotected during the most vulnerable phase—before payment is made.

Future of Anti-Scamming: Strategic Recommendations for Psyliq

1. **AI-Based Scam Language Detection**
   Real-time identification of high-risk scam narratives across platforms.
2. **Cross-Platform Behavioral Risk Scoring**
   Detection of scam patterns independent of communication channel.
3. **Cryptocurrency Transaction Signal Intelligence**
   Early identification of scam-linked wallets and laundering behavior.

Conclusion

This analysis demonstrates that modern scam operations are no longer opportunistic or isolated incidents, but organized systems

that adapt quickly to user behavior, technology, and enforcement gaps. The growing victimization of younger demographics, the increasing use of cryptocurrency for payments, and the reactive nature of current enforcement efforts highlight a clear need for earlier intervention.Effective scam prevention must shift from post-incident recovery to predictive and behavior-driven detection. By leveraging real-time language analysis, cross-platform intelligence, and transaction-level risk signals, Psyliq can play a critical role in disrupting scams before financial harm occurs.